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Inventor(s)	Volkov; Dmitry Aleksandrovich et al.

Method and system for detecting malicious infrastructure

Abstract

A method and a system for detecting a malicious infrastructure are provided. The method comprising: receiving a request comprising at least one infrastructure element of a given infrastructure assigned with a respective tag indicative of maliciousness of the given infrastructure; searching a database to identify therein, based at least on the respective tag, and at least one respective parameter of the at least one infrastructure element, and at least one additional infrastructure element and at least one additional respective parameter thereof; determining statistical relationships between the at least one respective parameter of at least one infrastructure element and the at least one additional respective parameter of the at least one additional infrastructure element; determining, based on the statistical relationships, rules for searching the database to identify therein new infrastructure elements; and assigning the respective tag associated with the given malicious infrastructure to the new infrastructure elements, thereby updating the database.

Inventors: **Volkov; Dmitry Aleksandrovich (Moscow, RU), Mileshin; Philipp Alekseevich (Moscow, RU)**

Applicant: **Group IB TDS, Ltd** (Moscow, RU)

Family ID: **1000008818769**

Assignee: **F.A.C.C.T. NETWORK SECURITY LLC (Moscow, RU)**

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Primary Examiner: Chen; Shin-Hon (Eric)

Attorney, Agent or Firm: BCF LLP

Background/Summary

CROSS-REFERENCE (1) The present application is a continuation of International Patent Application No. PCT/RU2020/000044, entitled “METHOD AND SYSTEM FOR MALWARE OR CYBERCRIMINAL INFRASTRUCTURE DETECTION,” filed on Jan. 31, 2020, the entirety of which is incorporated herein by reference.

FIELD
(1) This technical solution generally relates to the field of cybersecurity; and, in particular, to methods and systems for detecting malicious infrastructure.
BACKGROUND
(2) Certain prior art approaches for detecting malicious infrastructures have been proposed.
(3) Russian Patent No.: 2,523,114-C2 issued on Jul. 20, 2014 assigned to “Kaspersky Laboratory”, ZAO, and entitled “METHOD OF ANALYSING MALICIOUS ACTIVITY ON INTERNET,

DETECTING MALICIOUS NETWORK NODES AND NEIGHBOURING INTERMEDIATE NODES" discloses a method of detecting intermediate nodes in a computer network through which malware is distributed, wherein the intermediate nodes are connected to the Internet, to which malicious nodes are also connected. The present method employs a system of computer tools, services for detecting a traffic route in a network, a WHOIS service for accessing login information about an owner of a domain or IP address, followed by constructing flow chart of distribution of malware from a malicious site over data link channels, evaluating the usage rate of the link channel for distributing malware, detecting and blocking an intermediate node used illegally; the method further allows the unblocking of the intermediate node if the intensity of distribution of malware considerably drops over time or ceases to pose a threat to the site which directly contained the malware.

(4) U.S. Pat. No. 8,176,178-B2 issued on May 8, 2012, assigned to ThreatMetrix Pty Ltd, and entitled "METHOD FOR TRACKING MACHINES ON A NETWORK USING MULTIVARIABLE FINGERPRINTING OF PASSIVELY AVAILABLE INFORMATION" discloses a method for tracking machines on a network of computers. The method includes determining one or more assertions to be monitored by a first web site which is coupled to a network of computers. The method monitors traffic flowing to the web site through the network of computers and identifies the one or more assertions from the traffic coupled to the network of computers to determine a malicious host coupled to the network of computers. The method includes associating a first IP address and first hardware finger print to the assertions of the malicious host and storing information associated with the malicious host in one or more memories of a database. The method also includes identifying an unknown host from a second web site, determining a second IP address and second hardware finger print with the unknown host, and determining if the unknown host is the malicious host.

(5) Russian Patent No.: 2,634,209-C1 issued on Oct. 24, 2017, assigned to Group IB TDS, and entitled "SYSTEM AND METHOD OF AUTOGENERATION OF DECISION RULES FOR INTRUSION DETECTION SYSTEMS WITH FEEDBACK" discloses method includes the following steps: receiving at least one event from the event database generated by data received from at least one sensor; analyzing at least one received event for the class of interaction with the malware control centres; extracting from at least one of the above-mentioned events of the class of interaction with the malware control centres at least one feature used to form the decision rules; form decision rules using at least one of the above-mentioned extracted feature; storing the formed decision rules and providing an opportunity to receive an update of the decision rules for at least one sensor; sensors cyclically check the availability of updates in the central node and, if updates are available, receive them for use, and if updates are received, a trigger is activated in the sensors that restarts the decision rules.

SUMMARY

(6) It is an object of the present technology to ameliorate at least some of the inconveniencies of the prior art.

(7) Unlike the prior art, the present methods and systems allow identifying malicious infrastructures, such as those associated with a given piece of malware or with a cybercriminal, by determining certain rules, without analyzing traffic routes and the traffic itself. Each new rule enables to detect a respective malicious infrastructure. This includes, inter alia, identifying a currently operating malicious infrastructure, or the one which had worked a year ago and then was permanently deactivated, or even those malicious infrastructures which do not exist yet and are only being developed by a criminal-because he/she would disclose their preferences while developing a given future malicious infrastructure, which can further be identified by the rules disclosed herein.

(8) More specifically, in accordance with one broad aspect of the present technology, there is provided a computer-implementable method for detecting malicious infrastructure. The method is

executable by a processor communicatively coupled to an infrastructure element database. The method comprises: receiving, by the processor, a request comprising at least one infrastructure element of a given malicious infrastructure, the at least one infrastructure element being assigned with a respective tag indicative of maliciousness of the given malicious infrastructure; searching, by the processor, the infrastructure element database to identify therein, based at least on the respective tag associated with the at least one infrastructure element, at least one respective parameter thereof and at least one additional infrastructure element and at least one additional respective parameter thereof, the at least additional infrastructure element being assigned with the respective tag; analyzing, by the processor, (i) the at least one infrastructure element and the at least one respective parameter thereof, and (ii) the at least one additional infrastructure element and the at least one additional respective parameter thereof to determine statistical relationships between the at least one respective parameter of at least one infrastructure element and the at least one additional respective parameter of the at least one additional infrastructure element; determining, by the processor, based on the statistical relationships, rules for identifying, in the infrastructure element database, new infrastructure elements; retrieving, by the processors, using the rules, from the infrastructure element database, the new infrastructure elements; and assigning, by the processor, the respective tag associated with the given malicious infrastructure to the new infrastructure elements, thereby updating the infrastructure element database for further use in determining new infrastructure elements of the given malicious infrastructure.
(9) In some implementations of the method, the at least one infrastructure element of the given malicious infrastructure comprises at least one of: an IP address; a domain name; an SSL certificate; a server; a web service; an e-mail address; and a telephone number.
(10) In some implementations of the method, the infrastructure element database is continuously updated using data obtained by scanning the Internet.
(11) In some implementations of the method, the scanning the Internet is performed by at least one vulnerability scanner.
(12) In some implementations of the method, the respective tag is one of a conventional name and a common name of at least one of a piece malware and a cybercriminal associated with the given malicious infrastructure.
(13) In some implementations of the method, the request comprising the at least one infrastructure element is received from at least one of: a sandbox; a malware detonation platform; a vulnerability scanner; a honeypot; an intrusion detection system; and a system of emergency response to cyber security incidents.
(14) In some implementations of the method, the statistical relationships are determined using a trained machine learning algorithm.
(15) In some implementations of the method, the statistical relationships between the at least one respective parameter and at least one additional respective parameter represent a combination thereof by at least one logic operation, and wherein each of the at least one infrastructure element and the at least one additional infrastructure elements respectively associated therewith are different infrastructure elements.
(16) In some implementations of the method, the rules are stored and updated.
(17) In accordance with another broad aspect of the present technology, there is provided a system for malware or cybercriminal infrastructure detection. The system comprises: a processor communicatively coupled to an infrastructure element database and a non-transitory computer-readable memory storing instructions. The processor, upon executing the instructions, is configured to: receive a request comprising at least one infrastructure element of a given malicious infrastructure, the at least one infrastructure element being assigned with a respective tag indicative of maliciousness of the given malicious infrastructure; search the infrastructure element database to identify therein, based at least on the respective tag associated with the at least one infrastructure element, at least one respective parameter thereof and at least one additional infrastructure element

and at least one additional respective parameter thereof, the at least additional infrastructure element being assigned with the respective tag; analyze (i) the at least one infrastructure element and the at least one respective parameter thereof, and (ii) the at least one additional infrastructure element and the at least one additional respective parameter thereof to determine statistical relationships between the at least one respective parameter of at least one infrastructure element and the at least one additional respective parameter of the at least one additional infrastructure element; determine, based on the statistical relationships, rules for identifying, in the infrastructure element database, new infrastructure elements; retrieve, using the rules, from the infrastructure element database, the new infrastructure elements; and assign the respective tag associated with the given malicious infrastructure to the new infrastructure elements, thereby updating the infrastructure element database for further use in determining new infrastructure elements of the given malicious infrastructure. In the context of the present specification, the following terms should be interpreted as follows.

(18) Malicious infrastructure denotes a computer system running a piece of malware and/used by a cybercriminal for conducting malicious activity, and characterized, for example, without limitation, by at least some of: IP addresses, domain names, SSL certificates, servers (specially configured software) and web services, e-mail addresses, telephone numbers.

(19) Infrastructure element parameters denote parameters attributable to each infrastructure element of a given type. For example, the parameters characterizing such infrastructure element as web service may include a port number of a port where this web service is launched, date and time of its launch, and also date and time of its stop.

(20) Some parameters are specific to a particular type of infrastructure elements, for example, name of registrar company is specific to a parameter of “domain name” type. Other infrastructure elements could be more complex, that is, can be characterized by multiple parameters. For example, such specific parameter of IP address element as hosting provider is additionally characterized by geographic location, set of web services and servers available to hosting provider clients, range of IP addresses belonging to this hosting provider, etc.

(21) Statistical relationship between parameters of two and more infrastructure elements denotes at least one of an arithmetical, logical or other correspondence enabling to detect, for example, in a database, one more other infrastructure elements of a similar type, whose respective same-name parameters are linked to the parameters of the known infrastructure elements by the same relationship.

(22) For example, there are known IP addresses 11.22.33.44 and 55.66.77.88 issued by hosting provider Provider, LLC on the same day, 11.11.2011, to the client with contact address client@mail.com and contact phone+1 (099) 100-100-10. Statistical relationship between these two IP addresses and the third IP address 99.00.22.11, issued on some other day, for example, 12.12.2012, is that they are issued by the same hosting provider Provider, LLC to the client with the same contact address client@mail.com but with the other contact phone+1 (059) 200-200-20.

(23) Rule for searching for new infrastructure elements denotes a decision rule able to retrieve, in a database of a known structure, occurrences which satisfy the specified conditions, for example, all IP addresses issued by the specified hosting provider on a same day. The rule could be implemented as a script, regular expression, determined using a trained machine-learned algorithm, such as classifier, etc.

(24) Further, in the context of the present specification, unless expressly provided otherwise, a computer system may refer, but is not limited, to an “electronic device”, an “operation system”, a “system”, a “computer-based system”, a “controller unit”, a “control device” and/or any combination thereof appropriate to the relevant task at hand.

(25) In the context of the present specification, unless expressly provided otherwise, the expression “computer-readable medium” and “memory” are intended to include media of any nature and kind whatsoever, non-limiting examples of which include RAM, ROM, disks (CD-ROMs, DVDs,

floppy disks, hard disk drives, etc.), USB keys, flash memory cards, solid state-drives, and tape drives.
(26) In the context of the present specification, a "database" is any structured collection of data, irrespective of its particular structure, the database management software, or the computer hardware on which the data is stored, implemented or otherwise rendered available for use. A database may reside on the same hardware as the process that stores or makes use of the information stored in the database or it may reside on separate hardware, such as a dedicated server or plurality of servers.
(27) In the context of the present specification, unless expressly provided otherwise, the words "first", "second", "third", etc. have been used as adjectives only for the purpose of allowing for distinction between the nouns that they modify from one another, and not for the purpose of describing any particular relationship between those nouns.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Non-limiting embodiments of the present technology are described herein with reference to the accompanying drawings; these drawings are only presented to explain the essence of the technology and are not intended to limit the scope thereof in any way, where:
(2) FIG. **1** depicts a flowchart diagram of a method for detecting a malicious infrastructure, in accordance with certain non-limiting embodiments of the present technology; and
(3) FIG. **2** depicts a schematic diagram of an example computing environment configurable for execution of the present method of FIG. **1**, in accordance with certain non-limiting embodiments of the present technology.
DETAILED DESCRIPTION
(4) The following detailed description is provided to enable anyone skilled in the art to implement and use the non-limiting embodiments of the present technology. Specific details are provided merely for descriptive purposes and to give insights into the present technology, and in no way as a limitation. However, it would be apparent to a person skilled in the art that some of these specific details may not be necessary to implement certain non-limiting embodiments of the present technology. The descriptions of specific implementations are only provided as representative examples. Various modifications of these embodiments may become apparent to the person skilled in the art; the general principles defined in this document may be applied to other non-limiting embodiments and implementations without departing from the scope of the present technology.
(5) Certain non-limiting embodiments of the present technology are directed to methods and systems for detecting malicious infrastructures.
(6) Method
(7) With reference to FIG. **1**, there is depicted a flowchart diagram of a method (**100**) of detecting a malicious infrastructure, in accordance with certain non-limiting embodiments of the present technology. The method **100** can be implemented by a processor **201** of a computer system **200** described hereinbelow with reference to FIG. **2**.
(8) Step **101**: Receiving, by the Processor, a Request Comprising at Least One Infrastructure Element of a Given Malicious Infrastructure, the at Least One Infrastructure Element being Assigned with a Respective Tag Indicative of Maliciousness of the Given Malicious Infrastructure
(9) At step (**101**), according to certain non-limiting embodiments of the present technology, the processor **201** can be configured to receive a request comprising at least one infrastructure element of a given malicious infrastructure, and a respective tag of this element belonging to a malware or cybercriminal, that is, indicative of maliciousness of the given malicious infrastructure.
(10) In some non-limiting embodiments of the present technology, infrastructure elements can

comprise, for example, but without limitation, an IP address, a domain name, an SSL certificate, a server, a web service, an e-mail address, a telephone number. Conventional or common name of a malware or cybercriminal, for example, Zeus malware, could be used as the respective tag of a given infrastructure element belonging to a malware or cybercriminal.

(11) In some non-limiting embodiments of the present technology, the processor **201** can be configured to receive the request comprising the at least one infrastructure element from one of the following incoming request sources including, but not limited to, a sandbox (dedicated environment for program safe running) or a malware detonation platform, a vulnerability scanner, various honeypots (resource attracting the cybercriminal), an intrusion detection system, a system of emergency response to cyber security incidents.

(12) It is the vulnerability scanner that is used to take footprints from the services deployed at remote server ports. The vulnerability scanner as such could be implemented by any common method.

(13) All the listed programs and systems could be either inside one intranet with a system implementing the method, or at remote servers with Internet communication therebetween.

(14) The incoming request sources could also include systems of emergency response to cyber security incidents, CERT (Computer Emergency Response Team), automatically informing about a malicious activity in the monitored networks, remote computers, which owners send data on detection of malwares in their computers to the specified e-mail address, mobile phones, which owners send SMS or MMS containing data on malware running to the specified number.

(15) Further, in some non-limiting embodiments of the present technology, the incoming request sources could also include a system operator receiving the corresponding information by phone or e-mail.

(16) Further, the processor **201** can be configured to extract, from the request, information about the at least one infrastructure element, and also information about this element belonging to the given malicious infrastructure, that is, about the respective tag.

(17) For example, the sandbox has sent a request containing information that Zeus malware sends data from the infected computer to the server with the domain bad-domain.com, wherein this domain's current IP address is 11.22.33.44. The processor **201** can thus extract this IP address tagged with a respective Zeus tag.

(18) The method **100** thus advances to step **102**.

(19) Step **102**: Searching, by the Processor, the Infrastructure Element Database to Identify Therein, Based at Least on the Respective Tag Associated with the at Least One Infrastructure Element, at Least One Respective Parameter Thereof and at Least One Additional Infrastructure Element and at Least One Additional Respective Parameter Thereof, the at Least Additional Infrastructure Element being Assigned with the Respective Tag;

(20) Further, at step (**102**), according to certain non-limiting embodiments of the present technology, the processor **21** can be configured to extract, from a database, at least one respective parameter of the so obtained at least one infrastructure element, at least one additional infrastructure element of the given malicious infrastructure, and at least one additional respective parameter of the at least one additional infrastructure element. In some non-limiting embodiments of the present technology, the database may include data on infrastructure elements obtained from the Internet, such as by crawling, as an example.

(21) The processor **201** can be configured to search the database for all records comprising the at least one infrastructure element and also the respective tag associated therewith.

(22) Needless to mention that the database itself and the search in it could be implemented by any known method, for example, by that described in a Russian Patent No.: 2,681,699-C1 issued on Mar. 12, 2019, assigned to Trust LLC, and entitled “METHOD AND SERVER FOR SEARCHING RELATED NETWORK RESOURCES”, content of which is incorporated herein by reference in its entirety. Similarly to the method described in this patent, the database information could be

structured as a table illustrating relationship between at least the following parameters so that at least one of the following parameters corresponds to each infrastructure element, for example, domain name: tag, IP address, SSL key, SSH footprint, list of running services, history of running services, history of domain names, history of IP addresses, history of DNS servers, history of domain name or IP address owners.
(23) The parameters could also include DNS resource records which are records on compliance of name and service information in the domain name system.
(24) It should be noted that the data on the infrastructure elements could be stored in one or several databases or tables. In general, search is carried out by traditional methods: one or multiple request to these bases/tables and search for intersections of results.
(25) For example, as a result of a search for domain name bad-domain.com nothing was found in the database, during search for IP address 11.22.33.44 nothing was found in the database. However, by searching for the respective Zeus tag, the processor **201** can be configured to identify the IP address 55.66.77.88 also Zeus-tagged before.
(26) Thus, in accordance with certain non-limiting embodiments of the present technology, the processor **201** can be configured to retrieve the at least one additional infrastructure element and the at least one additional respective parameter associated therewith from the database for further analysis.
(27) In the above example, for IP address 55.66.77.88 found in the database there will be extracted, for example: domain name another-bad-domain.net, name of this IP address owner—Provider, LLC, name of this domain name registrar-Registrar, LLC, date of issue of this IP address-11.11.2011, date of registration of this domain name-22.12.2011, e-mail of the client to whom this IP address was issued-client@mail.com and client's contact phone number: +1 (099) 100-100-10.
(28) If some infrastructure elements are not found in the database at the second step, in some non-limiting embodiments of the present technology, the processor **201** can be configured to search online sources available for further retrieval therefrom of the parameters associated with such infrastructure elements. It could be done using, for example, a Whois online service or any known program fit for purpose, for example, an nslookup program.
(29) In this case, the processor **201** can be configured, for example, to automatically send the corresponding search request to the Whois online service and extract the required data from Whois online service response or from a web page with the search request results. The latter could be done by making use of, for example, the processor **201** executing a special parser and analyzing, for example, the text of Whois online service response or html code of the specified web page.
(30) In the given example for IP address 11.22.33.44 and domain name bad-domain.com, the processor **2021** can be configured to retrieve the following data: a name of this IP address owner-Provider, LLC, a name of this domain name registrar-AnotherRegistrar, LLC, a date of issue of this IP address-11.11.2011, a date of registration of this domain name-23.12.2011, an e-mail of the client to whom this IP address was issued-anotherclient@mail.com and a client's contact phone number: +1 (099) 300-300-30.
(31) Further, the processor **201** can be configured to store these parameters in the database together with the reference to the relationship between the said parameters, IP address 11.22.33.44, the domain name bad-domain.com, the respective Zeus tag, and also a date of retrieval of the specified infrastructure elements and a source from which they have been retrieved (in this case-identifier of the sandbox where Zeus malware activity was detected in the example in question).
(32) The method **100** hence proceeds to step (**103**).
(33) Step **103**: Analyzing, by the Processor, (I) the at Least One Infrastructure Element and the at Least One Respective Parameter Thereof, and (II) the at Least One Additional Infrastructure Element and the at Least One Additional Respective Parameter Thereof to Determine Statistical Relationships Between the at Least One Respective Parameter of at Least One Infrastructure Element and the at Least One Additional Respective Parameter of the at Least One Additional

Infrastructure Element
(34) At step (**103**), the processor **201** can be configured to analyze (i) the at least one infrastructure element and at least one respective parameter thereof, and (ii) the at least one additional infrastructure element and the at least one additional respective parameter thereof.
(35) Following the above example, the processor **201** can be configured to analyze the following infrastructure elements and their parameters: the domain names bad-domain.com and another-bad-domain.net, IP addresses 11.22.33.44 and 55.66.77.88, the names of this IP address owners (they are the same in our example), the names of this domain name registrars-Registrar, LLC and AnotherRegistrar, LLC, dates of issue of each IP address (they are the same in our example), the dates of registration of each domain name, 22.12.2011 and 23.12.2011, the client's e-mail addresses, the contact phone numbers; and the respective Zeus tag.
(36) Further, according to certain non-limiting embodiments of the present technology, the processor **201** can be configured to determine, based on the conducted analysis, statistical relationships between the at least one respective parameter of the at least one infrastructure element and the at least one additional respective parameter of the at least one additional infrastructure element.
(37) More specifically, the processor **201** can be configured to build the statistical relationships including arithmetical, logical, or other correspondence enabling to detect in the database, based on known relations between parameters of one-type infrastructure elements, at least one more new one-type infrastructure elements, which same-name parameters are linked to the parameters of the known elements with the same relation.
(38) For example, the processor **201** can be configured to determine that: the obtained IP addresses belong to one and the same owner, Provider, LLC, both IP addresses have been issued on the same day, 11.11.2011, client's e-mail addresses, although belong to different accounts, client@mail.com and anotherclient@mail.com, are registered at one and the same mail server mail.com.
(39) Correspondingly, the processor **201** can be configured to save the so determined statistical relationship in the following simple logical expression:
IPADDRESS_OWNER="Provider,LLC"_.Math._IPADDRESS_ISSUE_DATE="11.11.2011"_Λ_CLIENT_EMAIL="*@mail.com",
where Λ-conjunction operator, logical AND, *-so called wildcard character, which means that any number of any consecutive characters without spaces could be in its place.
(40) Hereinafter, statistical relationship features mean those infrastructure elements of parameters or infrastructure elements, a combination of which enables to build the detected statistical relationship. In the above example, this is the name of the IP addresses owner (provider), the IP address issue date and part of the client's e-mail address.
(41) It should be noted that the above examples is provided solely for clarity of explanation of the non-limiting embodiments of the present technology, and in more complex examples the processor **201** can be configured to receive for analysis a list of several hundreds of IP addresses registered at different times at different providers and not related to repeated or similar domain names, and also a list of running services interrelated with each of these IP addresses, and a list of running services history. When such data are analyzed by the processor **201**, more complex statistical relationship could be determined. For example, the processor **201** can be configured to determine that at each of these IP addresses Apache HTTP server was started at least once, which run for 48 hours and was stopped, and immediately after it stops the e-mail server CommuniGate Pro version 5.1 was started at the same IP address with XIMSS API interface support.
(42) Detection of statistical relationships described above could be performed by specifically trained machine learning algorithms.
(43) The method **100** thus proceeds to step (**104**).
(44) Step **104**: Determining, by the Processor, Based on the Statistical Relationships, Rules for Identifying, in the Infrastructure Element Database, New Infrastructure Elements

(45) At step (**105**), according to certain non-limiting embodiments of the present technology, the processor **201** can be configured to generate, based on the detected statistical relationship, rules for searching the database for new infrastructure elements.
(46) In the alternative non-limiting embodiments of the present technology, the processor **201** can be configured to generate the rules for searching for new infrastructure elements based on the data on the detected features of statistical relationships determined at step (**103**).
(47) A given rule is based on the set of features that cumulatively represent the detected statistical relationship determined at the previous step. According to certain non-limiting embodiments of the present technology, the given rule can include set of features expressed in some formal pseudolanguage. For example, a given formal pseudolanguage may include, without limitation, SQL, Python, and the like.
(48) Data, as stated above, could be stored in one or several bases/tables. Search is carried out by one or multiple requests to these bases/tables and search for intersections of results.
(49) If the given rule is generated for one feature, for example, for a footprint of a specific service, such as e-mail server CommuniGate Pro version 5.1, the search will be carried out in one base/table where only footprints of different services are stored, as an example.
(50) If the given rule is complex and comprises features of domains, certificates, etc., the requests will be submitted to different bases/tables, for searching for intersections of results.
(51) For example, according to a complex rule there could be carried out a search for all domains registered by AAAA registrar, wherein each of them (domains) has a certificate issued by BBBB company, and also each of the domains ever had IP address at which Apache HTTP had ever been started.
(52) The method **100** hence advances to step (**106**).
(53) Step **106**: Retrieving, by the Processors, Using the Rules, from the Infrastructure Element Database, the New Infrastructure Elements
(54) At step (**106**), in some non-limiting embodiments of the present technology, the processor **201** can be configured to retrieve, using the rule generated at the previous step, the new infrastructure elements from the database.
(55) To that end, the processor **201** can be configured to search for all records satisfying the conditions described by the rules.
(56) In the example provided above, as a result of the search in the database there could be identified a new IP address 99.00.22.11 without the respective Zeus tag but issued by the same hosting provider Provider, LLC to the client with the contact address thirdclient@mail.com on the same day 11.11.2011 as the previous two Zeus-tagged IP addresses.
(57) The method **100** hence advances to step (**107**).
(58) Step **107**: Assigning, by the Processor, the Respective Tag Associated with the Given Malicious Infrastructure to the New Infrastructure Elements, Thereby Updating the Infrastructure Element Database for Further Use in Determining New Infrastructure Elements of the Given Malicious Infrastructure
(59) At step (**107**), the processor **201** can be configured to assign the respective tag, such as the respective Zeus tag mentioned above, associated with the at least one infrastructure element and the at least one additional infrastructure element to the new infrastructure elements retrieved from the database using the rules determined as described above.
(60) More specifically, following the example provided above, the processor **201** can be configured to assign the respective Zeus tag to the found IP address and store the result in the database. By doing so, the processor **201** is configured to update the database for further use in identifying other/existing malicious infrastructures.
(61) It is worth noting that such algorithm allows to detect “sleeping”, i.e. already created but never used yet, infrastructure elements of malwares or cybercriminals.
(62) In the given example the found IP address 99.00.22.11 and related domain name could have

been created by developers of Zeus malware beforehand, however, they remain undetected by any conventional cybersecurity means as the malware has never contacted this server yet.
(63) Without direct reference to the above steps there is also continuous updating of the above-mentioned database. For example, the processor **201** can be configured to additionally update the database by continuous Internet scanning and taking footprints from the services launched at open server ports. At the same time, the processor **201** can be configured to collect data on domains related to the found servers.
(64) Taking footprints from the services launched at remote server ports can be carried out by the processor **201** executing a vulnerability scanner which could be implemented by any known method. The so identified results can further be arranged and stored, by the processor **201**, in the database.
(65) The method **100** thus terminates.
(66) Computer System
(67) With reference to FIG. **2**, there is depicted an example functional diagram of the computer system **200** configurable to implement certain non-limiting embodiments of the present technology including the method **100** described above.
(68) In some non-limiting embodiments of the present technology, the computer system **200** may include: the processor **201** comprising one or more central processing units (CPUs), at least one non-transitory computer-readable memory **202** (RAM), a storage **203**, input/output interfaces **204**, input/output means **205**, data communication means **206**.
(69) According to some non-limiting embodiments of the present technology, the processor **201** may be configured to execute specific program instructions the computations as required for the computer system **200** to function properly or to ensure the functioning of one or more of its components. The processor **201** may further be configured to execute specific machine-readable instructions stored in the at least one non-transitory computer-readable memory **202**, for example, those causing the computer system **200** to execute the method **100**.
(70) In some non-limiting embodiments of the present technology, the machine-readable instructions representative of software components of disclosed systems may be implemented using any programming language or scripts, such as C, C++, C#, Java, JavaScript, VBScript, Macromedia Cold Fusion, COBOL, Microsoft Active Server Pages, Assembly, Perl, PHP, AWK, Python, Visual Basic, SQL Stored Procedures, PL/SQL, any UNIX shell scrips or XML. Various algorithms are implemented with any combination of the data structures, objects, processes, procedures and other software elements.
(71) The at least one non-transitory computer-readable memory **202** may be implemented as RAM and contains the necessary program logic to provide the requisite functionality.
(72) The storage **203** may be implemented as at least one of an HDD drive, an SSD drive, a RAID array, a network storage, a flash memory, an optical drive (such as CD, DVD, MD, Blu-ray), etc. The storage **203** may be configured for long-term storage of various data, e.g., the aforementioned documents with user data sets, databases with the time intervals measured for each user, user IDs, etc.
(73) The input/output interfaces **204** may comprise various interfaces, such as at least one of USB, RS232, RJ45, LPT, COM, HDMI, PS/2, Lightning, FireWire, etc.
(74) The input/output means **205** may include at least one of a keyboard, a joystick, a (touchscreen) display, a projector, a touchpad, a mouse, a trackball, a stylus, speakers, a microphone, and the like. A communication link between each one of the input/output means **205** can be wired (for example, connecting the keyboard via a PS/2 or USB port on the chassis of the desktop PC) or wireless (for example, via a wireless link, e.g., radio link, to the base station which is directly connected to the PC, e.g., to a USB port).
(75) The data communication means **206** may be selected based on a particular implementation of a network, to which the computer system **200** can have access, and may comprise at least one of: an

Ethernet card, a WLAN/Wi-Fi adapter, a Bluetooth adapter, a BLE adapter, an NFC adapter, an IrDa, a RFID adapter, a GSM modem, and the like. As such, the input/output interfaces **204** may be configured for wired and wireless data transmission, via one of a WAN, a PAN, a LAN, an Intranet, the Internet, a WLAN, a WMAN, or a GSM network, as an example.

(76) These and other components of the computer system **200** may be linked together using a common data bus **210**.

(77) It should be expressly understood that not all technical effects mentioned herein need to be enjoyed in each and every embodiment of the present technology.

(78) Modifications and improvements to the above-described implementations of the present technology may become apparent to those skilled in the art. The foregoing description is intended to be provided as an example only rather than limiting. The scope of the present technology is therefore intended to be limited solely by the scope of the appended claims.

Claims

1. A computer-implementable method for detecting elements of a malicious infrastructure in an infrastructure element database, the method being executable by at least one processor, the method comprising: receiving a request comprising an infrastructure element of the malicious infrastructure, the malicious infrastructure being associated with a tag, the tag being indicative of at least one of a cybercriminal and a piece of malware known to be associated with the malicious infrastructure; the infrastructure element having a parameter, the infrastructure element comprising at least one selected from the group consisting of: an Internet Protocol (IP) address; a domain name; a Secure Sockets Layer (SSL) certificate; a server; a web service; an e-mail address; and a telephone number; and the parameter including at least one selected from the group consisting of: an other associated IP address, an SSL key, a Secure Shell (SSH) footprint, a list of running services, a history of running services, a history of domain names, a history of IP addresses, a history of Domain Name System (DNS) servers, a history of domain name or IP address owners, a hosting provider, and a geographic location; searching the infrastructure element database to identify therein, based at least on the tag associated with the infrastructure element, (i) an additional infrastructure element; (ii) and an additional parameter of the additional infrastructure element, the additional infrastructure element having been assigned with the tag, indicative of the additional infrastructure element being known to be associated with the malicious infrastructure; executing a machine-learning algorithm (MLA) to analyze (i) the infrastructure element and the parameter thereof, and (ii) the additional infrastructure element and the additional parameter thereof to determine whether the respective parameter of the infrastructure element and the additional parameter of the additional infrastructure element are both linked to at least one other parameter of a new infrastructure element, which is unidentified and unknown to be associated with any malicious infrastructure, the MLA having been trained to determine, based on previously determined relations of given-type parameters of a training infrastructure element with other parameters, whether the given-type parameters of the infrastructure element are linked to the other parameters according to the previously determined relations associated with the training infrastructure element; in response to determining that the respective parameter of the infrastructure element is linked to the at least one other parameter akin to the additional parameter of the additional infrastructure element being linked to the at least one other parameter, generating rules for identifying, in the infrastructure element database, new infrastructure elements, which were previously unidentified and unknown to be associated with any malicious infrastructure, a given rule of the rules including a regular expression that is representative of how the respective parameter and the additional respective parameter are linked to a given one of the at least one other parameter; retrieving, using the rules, from the infrastructure element database, the new infrastructure elements; and determining the new infrastructure elements as being associated with

the malicious infrastructure by assigning the tag to the new infrastructure elements, thereby updating the infrastructure element database for further use in determining other new infrastructure elements of the malicious infrastructure.
2. The method of claim 1, wherein the infrastructure element database is continuously updated using data obtained by scanning the Internet.
3. The method of claim 2, wherein the scanning the Internet is performed by at least one vulnerability scanner.
4. The method of claim 1, wherein the tag is one of a conventional name and a common name of the at least one selected from the group consisting of the piece malware and the cybercriminal associated with the malicious infrastructure.
5. The method of claim 1, wherein the request comprising the at least one infrastructure element is received from at least one of: a sandbox; a malware detonation platform; a vulnerability scanner; a honeypot; an intrusion detection system; and a system of emergency response to cyber security incidents.
6. The method of claim 1, wherein the determining whether the parameter and the additional parameter are both linked to the at least one other parameter comprises determining a combination thereof by at least one logic operation, and wherein each of the infrastructure element and the additional infrastructure elements respectively associated therewith are different infrastructure elements.
7. The method of claim 1, wherein the rules are stored and updated.
8. A system for detecting elements of a malicious infrastructure in an infrastructure element database, the system comprising: at least one processor communicatively coupled to the infrastructure element database; at least one non-transitory computer-readable memory storing executable instructions, which, when executed by the at least one processor, cause the system to: receive a request comprising an infrastructure element of the malicious infrastructure, the malicious infrastructure being associated with a tag, the tag being indicative of at least one of a cybercriminal and a piece of malware known to be associated with the malicious infrastructure; the infrastructure element having a parameter, the infrastructure element comprising at least one selected from the group consisting of: an Internet Protocol (IP) address; a domain name; a Secure Sockets Layer (SSL) certificate; a server; a web service; an e-mail address; and a telephone number; and the parameter including at least one selected from the group consisting of: an other associated IP address, an SSL key, a Secure Shell (SSH) footprint, a list of running services, a history of running services, a history of domain names, a history of IP addresses, a history of Domain Name System (DNS) servers, a history of domain name or IP address owners, a hosting provider, and a geographic location; search the infrastructure element database to identify therein, based at least on the tag associated with the infrastructure element, (i) an additional infrastructure element; (ii) and an additional parameter of the additional infrastructure element, the additional infrastructure element having been assigned with the tag, indicative of the additional infrastructure element being known to be associated with the malicious infrastructure; execute a machine-learning algorithm (MLA) to analyze (i) the infrastructure element and the parameter thereof, and (ii) the additional infrastructure element and the additional parameter thereof to determine whether the parameter of the infrastructure element and the additional parameter of the additional infrastructure element are both linked to at least one other parameter of a new infrastructure element, which is unidentified and unknown to be associated with any malicious infrastructure, the MLA having been trained to determine, based on previously determined relations of given-type parameters of a training infrastructure element with other parameters, whether the given-type parameters of the infrastructure element are linked to the other parameters according to the previously determined relations associated with the training infrastructure element; in response to determining that the respective parameter of the infrastructure element is linked to the at least one other parameter akin to the additional parameter of the additional infrastructure element being linked to the at least one

other parameter, generate rules for identifying, in the infrastructure element database, new infrastructure elements, which were previously unidentified and unknown to be associated with any malicious infrastructure, a given rule of the rules including a regular expression that is representative of how the respective parameter and the additional respective parameter are linked to a given one of the at least one other parameter; retrieve, using the rules, from the infrastructure element database, the new infrastructure elements; and determine the new infrastructure elements as being associated with the malicious infrastructure by assigning the tag to the new infrastructure elements, thereby updating the infrastructure element database for further use in determining other new infrastructure elements of the malicious infrastructure.